

Programme	B. Sc. STATISTICS				
Course Code	STA 8 EJ 414				
Course Title	STATISTICAL DECISION THEORY				
Type of Course	Major Elective				
Semester	VIII				
Academic Level	400-499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	Statistical testing hypothesis, Priori & Posterior probability				
Course Summary	To understand different decision rule using statistics and Bayesian analysis .				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
1.	Explain different loss functions and decision principle	Ap	C	Seminar Presentation /
2.	Describe the use of prior information in decision making.	R	F	Instructor-created exams
3.	Calculate Posterior distribution and check the admissibility of Bayes rules	Ap	C	Group Tutorial Work
4.	Develop general techniques for solving games	Ap	C	Home Assignments
* - Remember (R), Understand (U), Apply (Ap), Analyze (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (60)	Marks (70)
I	Statistical decision Problem		12	15
	1	Decision rule		
	2	Loss-randomized decision rule		
	3	Decision Principle - sufficient statistic and convexity		
	4	Utility and		
	5	Loss-loss functions		
	6	Standard loss functions vector valued loss functions		

II	Prior information		12	20
	7	subjective determination of prior density		
	8	Non-informative priors		
	9	Maximum entropy priors he marginal distribution to determine the prior		
	10	the ML-II approach to prior selection		
	11	Conjugate priors		
III	The posterior distribution		12	20
	12	Bayesian inference		
	13	Bayesian decision theory		
	14	Empirical Bayes analysis		
	15	Hierarchical Bayes analysis		
	16	Bayesian robustness Admissibility of Bayes rules		
IV	Game theory		12	15
	17	Basic concepts		
	18	General techniques for solving games		
	19	Games with finite state of nature		
	20	the supporting and separating hyper plane theorems		
	21	The minimax theorem		
	22	Statistical games		
V	Open –Ended		12	
	Problems regarding Module I to Module IV			

Text Book

Berger, O.J. (1985). Statistical Decision Theory and Bayesian Analysis – Second Edition. Springer, New York.

Reference

- 1. Ferguson, T.S. (1967).** Mathematical Statistics-A Decision Theoretic Approach. Academic Press, New York.
- 2. Lehman, E.L. (1998).** Theory of Point Estimation-Second Edition. John Wiley, New York.